**Did They Grow Up With the Game? Directors' Age and Critics' Scores for Films Based on Video Games, 1993–2025**

Course example · ECON 3916 Midterm Project · Dr. Richeng Piao

**Repository:** github.com/RyanPiao/econ3916-midterm-example **Notebook:** colab.research.google.com/github/RyanPiao/econ3916-midterm-example/blob/main/midterm-example-movies.ipynb

| | |
|---|---|
| **Question** | Among English-language films based on video games, 1993–2025, are films whose directors were younger when the game came out rated better by critics (the Rotten Tomatoes score), holding the release year and the format constant? |
| **Hypothesis** | A director who was young when the game came out is more likely to have grown up playing it, so knows what its players care about and should make a better film of it: a negative coefficient on that age. |
| **Headline finding** | A director who was ten years younger when the game came out goes with a score 2.9 points higher (95% CI −2.3 to +8.0 points, N = 53). Before the controls the association is clear; holding the release year fixed shrinks it to an estimate whose interval includes zero, and a median regression gives the opposite sign. The hypothesis is neither supported nor rejected. |
| **Data** | One row is one film (53 in the sample, from 68 listed). Scores: Wikipedia's "List of films based on video games", citing Rotten Tomatoes, downloaded 2026-11-03. When each game came out: the first release year on the Wikipedia page of the game or series each film is based on. Directors' birth dates: Wikidata. |
| **Method** | OLS of the Rotten Tomatoes score (percent of critics' reviews that are positive) on the director's age when the game came out, the release year and an animated indicator, with robust (HC1) standard errors. |

## 1. The Question and the Data

Video-game adaptations have a famously bad record with critics, and for years the usual explanation was that the people making them did not know the games. That is changing. Many of the directors hired in the last decade were children or teenagers when the games they adapt first came out. If a director who grew up with a game makes a better film of it, a studio choosing a director has a reason to look for one. The question matters to the studios making that choice and to the players who wait for these films.

My question is whether, among English-language films based on video games released from 1993 to 2025, films whose directors were younger when the game came out are rated better by critics, holding the release year and the format constant. One row is one film. The outcome is the film's Rotten Tomatoes score: the percent of critics' reviews that are positive. The main explanatory variable is the director's age when the game came out: the first release year of the game or series the film is based on, minus the director's birth year. A value of 10 means the director was ten when the game appeared. My hypothesis is that a director who was young when the game came out is more likely to have played it, so I expect a negative coefficient. I chose this X over the director's age when the film was made because it

matches the mechanism: two directors of the same age can be adapting a game that came out in their childhood or one that came out last year.

The data are a product of several measurement decisions. The list of films comes from Wikipedia's "List of films based on video games", which volunteer editors keep up to date and which cites Rotten Tomatoes for every score. Rotten Tomatoes decides which critics count and whether each review is positive or negative, so a score of 30% means that three in ten counted reviews were positive, not that the film earned 30 out of 100. The score measures critics, not players, and it is noisier when few critics reviewed the film. When each game came out comes from Wikipedia too: each film's infobox names the game or series it is "based on", and that page gives the first release year. Directors' birth dates come from Wikidata. Budgets, used only in Section 3, are put in 2024 prices with the BLS consumer price index (CPIAUCSL) from FRED.

Building the sample took four steps and two joins, and each changed the data. The page lists 68 films, 55 live-action and 13 animated. The title is not a key: Mortal Kombat, Resident Evil and Street Fighter were each remade, so I identify a film by its title and release year, and that key matters again when the game years are joined: the 1994 Street Fighter is based on Street Fighter II (1991), the 2026 one on the 1987 original. The scores arrive as text ("26% (46 reviews)[4]"), which I split into the score and the number of reviews. Merging the films with the birth dates on the director's name first matched only 58 of 68 films, because films with two directors came out of the web table with the names glued together ("Rocky MortonAnnabel Jankel"); matching on the first director listed fixed that. The game years matched 66 of the 68 films.

Two gaps matter. 4 films have no score (Far Cry, Dead Trigger, Heavenly Sword, Scarygirl). Rotten Tomatoes posts no score below a minimum number of reviews, so whether a score exists depends on the review count, which I observe: the gap is not MCAR, and MAR given the review count is the defensible label. A film too weak to draw critics would also land here, which would make it MNAR in the score itself; that is possible and cannot be tested. Its effect is that the sample is the films critics bothered to review. Second, 5 rated films lose the director's age because Wikidata has no birth date. Their median score is 43.0%, against 29.0% for the rest, so this gap is not MCAR either; 2 of them are animated, against 15% of the sample, so with only five films I treat it as MAR given the format, which the model controls for. Films released in 2026 are dropped by a rule written before fitting any model, since their reviews are still coming in. Table 1 shows each step; Table 2 summarizes the 53 films that remain.

**Table 1. From 68 listed films to the sample**

| Step | Films |
|---|---|
| Films listed | 68 |
| Released in 2026 (reviews still coming in) | −6 |
| No Rotten Tomatoes score | −4 |
| No birth date for the director | −5 |
| No release year for the source game | 0 |

| | |
|---|---|
| Films in the sample | 53 |

Steps are applied in order, so a film missing two things is counted once, at the first step it fails.

**Table 2. Summary statistics, 53 films, 1993–2025**

| Variable (units) | Mean | Median | SD | Min | Max | N |
|---|---|---|---|---|---|---|
| Rotten Tomatoes score (% positive reviews) | 34.2 | 29.0 | 24.5 | 1.0 | 100.0 | 53 |
| Reviews counted | 134.3 | 120.0 | 83.0 | 5.0 | 327.0 | 53 |
| Director's age when the game came out (years) | 31.2 | 32.0 | 10.8 | 1.0 | 53.0 | 53 |
| Director's age at release (years) | 45.0 | 44.0 | 7.7 | 30.0 | 68.0 | 53 |
| Release year | 2011.8 | 2012 | 8.9 | 1993 | 2025 | 53 |
| Budget ($m, 2024) | 86.5 | 74.4 | 59.2 | 7.5 | 251.7 | 50 |

Budget in millions of 2024 dollars (CPI-U); 50 of the 53 films report one in dollars. Sources: Wikipedia (citing Rotten Tomatoes, Box Office Mojo and The Numbers), Wikidata, FRED. Downloaded 2026-11-03.

## 2. Descriptive Evidence

Most of these films are reviewed badly. The median score is 29.0%, the mean 34.2%, and only 9 of the 53 films pass 60% (Figure 1). The distribution is skewed right (skewness 0.75), with a long tail of well-reviewed films, which is why Table 2 reports the median beside the mean. A score is already a percentage between 0 and 100, so I model it directly rather than in logs, and check the result with a median regression in Section 4.

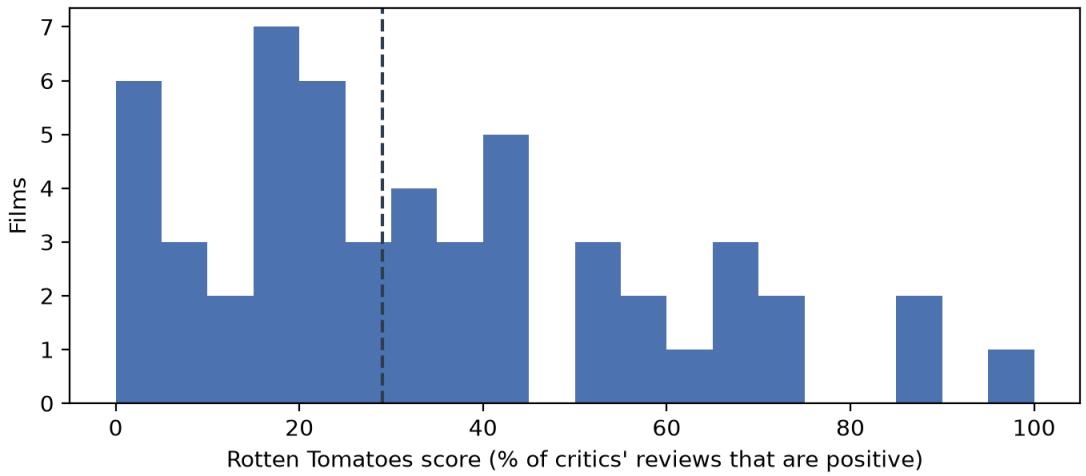



**Figure 1.** Most video-game adaptations are reviewed badly: the median film (dashed line) has 29.0% positive reviews, and only 9 of the 53 pass 60%. Source: Rotten Tomatoes scores as cited by Wikipedia.

Figure 2 plots each film's score against the director's age when the game came out, which runs from 1 to 53 years. The cloud tilts down: films by directors who were younger when the game appeared are rated better. Figure 3 shows why that raw pattern cannot be read at face value. Scores rise with the release

year: films released from 2016 on average 48.3%, against 20.5% for 1993–2005 and 25.5% for 2006–2015. And the recent films adapt games that came out when their directors were younger: the director's age when the game came out averages 34.5 years for films released before 2016 and 26.8 for films from 2016 (correlation with the release year -0.33). Part of the tilt in Figure 2 is the release year.




**Figure 2.** Films by directors who were younger when the game came out are rated better, before any controls; the youngest directors were about 1 when their game appeared. Source: as Figure 1; game years from Wikipedia, birth dates from Wikidata.

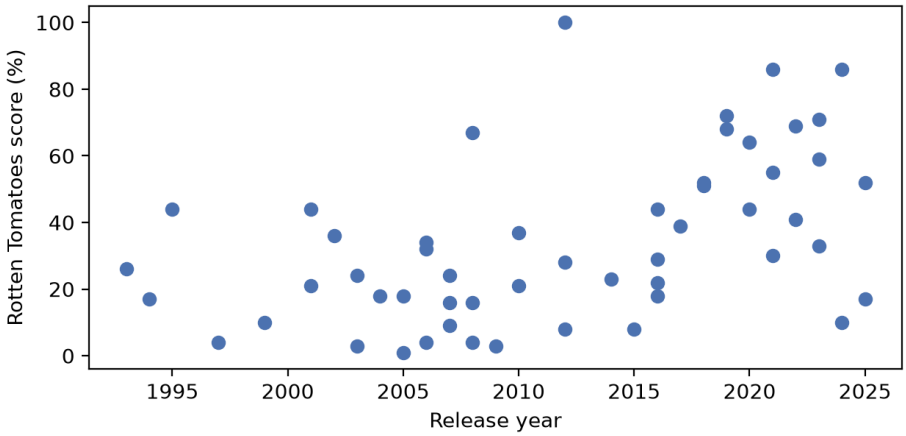



**Figure 3.** Recent adaptations are rated far better: films from 2016 on average 48.3%, against 20.5% to 25.5% before. Source: as Figure 1.

## 3. The Model

I fit the model below by OLS, with heteroskedasticity-robust (HC1) standard errors:

$$Score_i = \beta_0 + \beta_1\, AgeWhenGameCameOut_i + \beta_2\, Year_i + \beta_3\, Animated_i + u_i$$

The full results are in Table A1 in the appendix; column 2 is the main model. Without any controls (column 1) the coefficient on age when the game came out is −0.71 points per year (95% CI −1.16 to −0.26), an interval that excludes zero. Holding the release year and the format fixed (column 2), it is

−0.29 points per year (standard error 0.26): a director who was ten years younger when the game came out goes with a score 2.9 points higher, with a 95% confidence interval from −2.3 to +8.0 points. The point estimate keeps the sign my hypothesis predicts, but the interval now includes zero and a small effect the other way. With 53 films the data do not decide.

The second coefficient I interpret is the release year: +1.07 points per year (95% CI +0.40 to +1.74). Holding the director's age when the game came out and the format fixed, a film released ten years later scores about 11 points higher, and this interval excludes zero. The animated indicator is +25.2 points (95% CI +7.0 to +43.3), though only 8 films are animated. The $R^2$ is 0.39, against 0.26 without the release year.

Figure 4 draws the assumptions behind the model, and each control has a role in it:

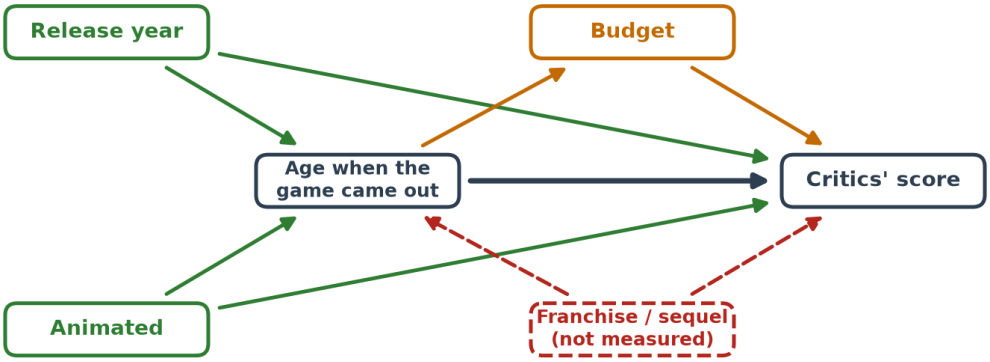



**Figure 4.** The causal graph I assume. Release year and format point into both the director's age when the game came out and the score, so the model controls for them. Budget could sit between the director and the score, so it stays out. Franchise and sequel status, which I cannot measure, leaves a path open (Section 5).

- **Release year** is a confounder, and here it does real work. Scores rose sharply over three decades (Figure 3), and recent films adapt games that came out when their directors were younger (Section 2). Leaving it out credits the director with the improvement of the whole genre: it is why the coefficient shrinks from −0.71 to −0.29. It would be a bad control only if the director's age when the game came out could change the film's release year, which it cannot.
- **Animated** also blocks a confounding path: animated adaptations are made by different directors and reviewed on a different curve. The studio picks the format before hiring the director, so the director's age cannot cause it.
- **Budget** is left out on purpose. A studio may give a bigger budget to a director it trusts to know the game, and the money can change how good a film is, so budget would be a mediator and a bad control for my question. In these films it barely matters: the director's age when the game came out is almost unrelated to the budget (−0.2% per year, holding year and format fixed), and columns 3 and 4, on the same 50 films, give −0.34 and −0.33. The number of reviews is not a control either, because it depends on how much attention the film drew.

## 4. Diagnostics

I checked the main model with the tests from Topic 14. The residual plots in Figure A1 show no curve against the fitted values or against age when the game came out, so a straight line is a reasonable

shape. The spread of residuals is wide, a standard deviation of 19.7 points: critics' scores vary a great deal around the model.

Constant variance: the Breusch–Pagan test gives p = 0.28, no evidence of unequal variance. With 53 films the test has little power, so I lean on the plots and report robust standard errors anyway. The choice is visible in the age coefficient: its standard error is 0.27 with classical errors (95% CI −0.83 to +0.25) and 0.26 with HC1 (−0.80 to +0.23). Both include zero, so the conclusion does not depend on it.

Collinearity: the variance inflation factors are 1.14 for age when the game came out, 1.13 for the release year and 1.04 for the animated indicator. Age and year are related (Section 2), but not enough to inflate the standard errors much; the wide interval comes from the small sample and the noisy scores.

Influence: using Cook's distance with the cutoff 4/n = 0.075, 7 films stand out. The three largest are Resident Evil: Damnation (0.214), Ratchet & Clank (0.160) and Sonic the Hedgehog 3 (0.111). Resident Evil: Damnation's score of 100% rests on 5 reviews. A score is a share of reviews, so its precision rises with the review count, and a share from five critics is mostly noise. Without the 3 films reviewed fewer than 20 times (Resident Evil: Degeneration, Resident Evil: Damnation, Resident Evil: Death Island), the coefficient is −0.34 points per year (standard error 0.25, N = 50).

Three more checks. A median regression, because the score is skewed, gives +0.21 points per year (95% CI −0.49 to +0.92): the point estimate changes sign, and the interval includes zero. Using the director's age at release instead, the cruder X, gives −0.17 (95% CI −0.81 to +0.46). And a threshold in place of a line: films whose directors were under 20 when the game came out score 24.1 points higher (95% CI +7.6 to +40.6), but that group is only 6 films (Rampage, Sonic the Hedgehog, Mortal Kombat, Sonic the Hedgehog 2, Sonic the Hedgehog 3, The Super Mario Bros. Movie), all recent franchise films, three of them Sonic. The large number is real in the data and impossible to separate from franchise and timing.

## 5. Threats and Conclusion

For the coefficient to be causal, nothing else that moves a film's score could vary with the director's age when the game came out once the year and the format are fixed. Franchises break that. The films whose directors grew up with the game are recent entries in big, carefully managed series (Sonic, Mortal Kombat, Mario), and studios hand those to directors they already trust; my data have no franchise or sequel variable to block that path. Measurement is the second threat. Age when the game came out is still a stand-in for having played it: a director of 12 may never have owned a console, and one of 35 may have played it every night. The "based on" page also decides the date: the 2018 Tomb Raider is based on the 2013 reboot, not the 1996 original. And the score measures critics, not the players the hypothesis is about.

So the estimate is a conditional association, not the effect of having grown up with the game. Among these 53 films, holding the release year and the format fixed, a director who was ten years younger when the game came out goes with a score 2.9 points higher, with an interval from −2.3 to +8.0 points. The hypothesis is neither supported nor rejected: the raw association is clear, most of it goes with the release year, and what is left has the predicted sign in OLS, the opposite sign in a median regression, and intervals that include zero. With another month I would code each director's own gaming history from

interviews, add a franchise flag, weight each score by its review count, and add audience scores to see whether players and critics agree.

## References

Federal Reserve Bank of St. Louis. Consumer Price Index for All Urban Consumers: All Items in U.S. City Average (CPIAUCSL), from the U.S. Bureau of Labor Statistics. FRED. Retrieved 2026-11-03.

Wikidata. Date of birth (P569) for each director, retrieved through the Wikidata API on 2026-11-03.

Wikipedia contributors. “List of films based on video games.” Wikipedia. Retrieved 2026-11-03. Scores as cited there from Rotten Tomatoes; budgets from Box Office Mojo and The Numbers. Game release years from the Wikipedia page of the game or series each film’s infobox names, retrieved 2026-11-04.

## Appendix

**Table A1. Rotten Tomatoes score regressed on the director's age when the game came out and controls**

| | (1) Age only | (2) Main | (3) Main, budget films | (4) Adds budget |
|---|---|---|---|---|
| Age when the game came out (years) | -0.71 | -0.29 | -0.34 | -0.33 |
| | (0.23) | (0.26) | (0.25) | (0.24) |
| Release year | | 1.07 | 1.11 | 1.14 |
| | | (0.34) | (0.36) | (0.36) |
| Animated (0/1) | | 25.18 | 12.16 | 10.83 |
| | | (9.27) | (8.51) | (7.79) |
| log budget | | | | 4.54 |
| | | | | (5.34) |
| Constant | 56.37 | -2119.11 | -2188.28 | -2269.63 |
| | (7.91) | (688.86) | (721.00) | (732.41) |
| N | 53 | 53 | 50 | 50 |
| $R^2$ | 0.10 | 0.39 | 0.34 | 0.36 |

Note: Dependent variable: Rotten Tomatoes score, percent of critics' reviews that are positive. OLS with robust (HC1) standard errors in parentheses. Column 2 is the main model. Column 3 re-estimates it on the 50 films with a dollar budget, so that column 4, which adds the log budget (a possible mediator, Section 3), compares the same films.

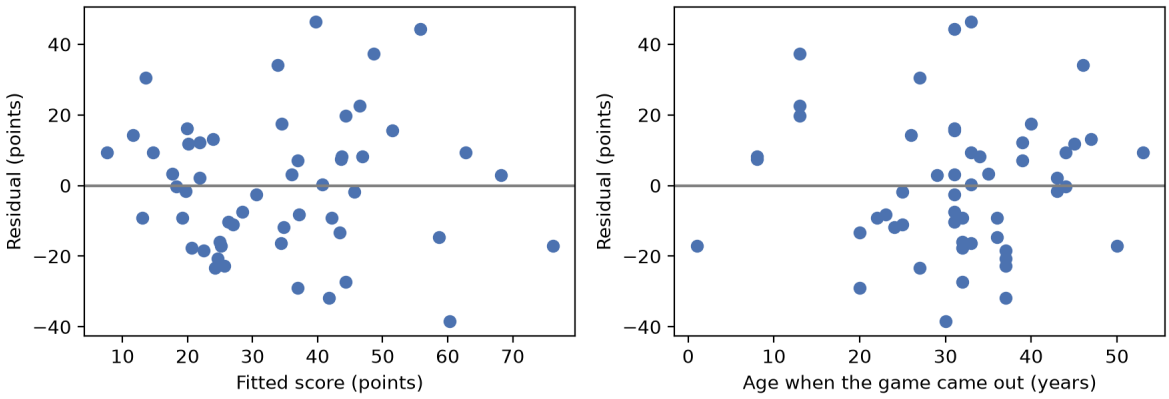



**Figure A1.** The main model's residuals show no curve against the fitted values (left) or the director's age when the game came out (right); the spread is wide throughout.

**Reproducing the numbers.** Every number in this report is printed by the last code cell of the notebook in the repository, "Numbers my report quotes". The README gives the download step and the order to run the cells.